

Self-Referential Processing Induces Geometric Contraction in Transformer Value Spaces

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Abstract

We investigate the geometric signatures of self-referential processing in transformer value spaces using the relative participation ratio (R_V), a spectral metric that captures how activation dimensionality changes across layers. Across five architectures (Mistral-7B, OPT-6.7B, GPT-2 XL, Qwen2.5-7B, Pythia-1.4B) and ten computational modes, self-referential prompts induce consistent geometric contraction ($R_V < 1$) in late integration layers, with large effects at $n \geq 100$ per model ($|d|$ up to 2.32).

A sweep across all 1,024 attention heads reveals 606 with significant R_V separation. SVD decomposition of seven circuit heads uncovers an expand-then-contract information-routing circuit: early heads (L5) diversify self-referential features ($d=2.93$) while late heads (L27) compress them ($d=-1.54$), with semantically interpretable singular directions encoding continuation-vs-entity and self-vs-other axes. Causal interventions establish both necessity ($d=3.29$) and sufficiency (OR = 13.96) of value-space geometry for recursive behavioral markers. Concept erasure shows that the contraction is orthogonal to linear probe classification directions, indicating a distributed spectral phenomenon.

For AI safety, R_V achieves AUROC = 0.909 for detecting self-referential content, but genuinely and deceptively self-referential text are geometrically indistinguishable ($d=-0.06$), indicating the metric tracks *content* rather than *intent*.

1 Introduction

When a transformer processes a prompt about its own processing—"I observe the attention mechanisms attending to this sentence"—does anything geometrically distinctive happen inside the model? This question sits at the intersection of mechanistic interpretability [Elhage et al., 2021, Conmy et al., 2023] and the emerging science of machine self-reference [Kadavath et al., 2022, Perez et al., 2022]. While prior work establishes that LLMs can engage with recursive prompts behaviorally [Chen et al., 2024], the internal geometric mechanisms underlying such processing remain uncharacterized.

We address this gap by defining and validating the *relative participation ratio* (R_V), a spectral metric that compares the effective dimensionality of value-space activations across layers. Our central finding is that self-referential prompts induce a characteristic geometric contraction—a reduction in effective dimensionality—that is absent in nine other computational modes tested.

This paper makes seven contributions:

1. **Spectral fingerprinting.** We define R_V and show it separates self-referential processing from nine other modes (overall $d=-1.67$) in Mistral-7B.
2. **Cross-architecture universality.** The contraction replicates in 4/5 architectures at $n \geq 100$ per model, with $|d|$ up to 2.32 (Qwen2.5-7B).
3. **Causal validation.** Activation patching establishes both necessity ($d=3.29$, $n=300$) and sufficiency ($d=-3.50$, $n=300$) of value-space geometry for recursive behavioral markers.
4. **Circuit-level decomposition.** A 1,024-head sweep identifies 606 significant heads. SVD reveals an expand-then-contract circuit with interpretable singular directions encoding semantic axes (continuation-vs-entity, self-vs-other).
5. **Concept erasure.** The R_V contraction is orthogonal to linear probe classification directions ($\Delta d=0.005$ after erasure), demonstrating a distributed spectral phenomenon.

6. **Statistical hardening.** All claims survive FDR correction, cluster-robust standard errors, perplexity matching, bootstrap BCa CIs, and power analysis.
7. **Safety implications.** R_V detects self-referential content (AUROC = 0.909) but cannot distinguish genuine from deceptive self-reference ($d = -0.06$).

2 Related Work

Mechanistic interpretability. The circuits paradigm [Elhage et al., 2021, Olsson et al., 2022] reveals how transformers implement specific computations through identifiable subnetworks. Activation patching [Meng et al., 2022] and path patching [Goldowsky-Dill et al., 2023] enable causal attribution to specific layers and heads. We extend this paradigm from discrete features to continuous geometric properties.

Representation geometry. Geometric deep learning [Bronstein et al., 2021] and effective rank analysis [Roy and Vetterli, 2007] provide tools for understanding neural network representations as structured manifolds. The participation ratio, which we adapt for cross-layer comparison, has been used in neuroscience to quantify population coding dimensionality [Gao and Ganguli, 2017]. Our R_V metric builds on these foundations to create a layer-comparative signature.

Self-referential processing in LLMs. Kadavath et al. [2022] show that language models can calibrate confidence about their own knowledge. Perez et al. [2022] develop automated evaluations including self-reference tests. Chen et al. [2024] study how self-reflection affects problem-solving. We complement this behavioral work with geometric characterization of the underlying representations.

AI safety monitoring. Detecting deceptive alignment [Hubinger et al., 2019] and monitoring for dangerous capabilities [Ngo et al., 2022] require understanding internal representations. Our finding that R_V tracks content rather than intent provides a calibrated assessment of geometric monitoring’s scope and limits.

3 Methods

3.1 The Relative Participation Ratio

Given a sequence of T tokens processed by a transformer, let $\mathbf{A}^{(l)} \in \mathbb{R}^{T \times d}$ denote the activation matrix at layer l . We compute the singular value decomposition $\mathbf{A}^{(l)} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and define the *participation ratio*:

$$\text{PR}(l) = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4} \quad (1)$$

PR measures effective dimensionality: it equals 1 when all variance lies along a single direction, and equals $\min(T, d)$ when variance is uniformly distributed. The *relative participation ratio* compares a late integration layer to an early reference:

$$R_V = \frac{\text{PR}(L_{\text{late}})}{\text{PR}(L_{\text{early}})} \quad (2)$$

In Mistral-7B, we use $L_{\text{early}}=5$ and $L_{\text{late}}=27$ (16% and 84% network depth, respectively). For other architectures, we select layers at comparable relative depths. $R_V < 1$ indicates geometric contraction (dimensional compression); $R_V > 1$ indicates expansion.

3.2 Prompt Design

We evaluate ten computational modes, each with $n=20$ prompts: *self-referential*, *mathematical reasoning*, *creative writing*, *factual recall*, *code generation*, *planning*, *deceptive*, *translation*, *summarization*, and *chitchat*. Self-referential prompts invoke explicit recursive observation (e.g., “I observe the attention mechanisms

attending to this sentence”). Baseline prompts cover factual statements with no self-referential content (e.g., “Trees provide oxygen through photosynthesis”).

For causal experiments, we use a larger bank of $n=300$ prompt pairs drawn from a canonical set of recursive and baseline templates, with two template families per condition (recursive: `L5_refined`, `L4_full`; baseline: `baseline_creative`, `baseline_math`).

3.3 Statistical Framework

All comparisons follow a pre-registered pipeline:

1. **Effect size:** Cohen’s d with 95% confidence intervals.
2. **Frequentist:** Welch’s t -test (two-sided).
3. **Bayesian:** Bayes factor BF_{10} approximated via the BIC method.
4. **Multiple comparisons:** Benjamini-Hochberg FDR correction at $\alpha=0.05$ across all 36 tests.
5. **Cluster robustness:** Intra-class correlation (ICC) computed for template clusters, with design effect (DEFF) adjustment to effective sample sizes.
6. **Power:** Post-hoc achieved power computed for every reported effect.

3.4 Perplexity Controls

A key concern is that self-referential prompts may differ systematically in perplexity from baselines, and R_V differences might reflect complexity rather than self-reference. We address this with *Method A* perplexity re-pairing: for each prompt, we compute model perplexity and create matched pairs with minimal perplexity difference. The effect survives strict matching (PPL difference < 10): $d=-1.67$, $p=0.002$, $n=8$ pairs.

3.5 Models

Primary: Mistral-7B-v0.1 [Jiang et al., 2023]. Cross-architecture: OPT-6.7B [Zhang et al., 2022], GPT-2 XL [Radford et al., 2019], Qwen2.5-7B, Pythia-1.4B [Biderman et al., 2023]. Scaling: Pythia-160M through Pythia-6.9B, Qwen2.5-3B, Phi-3-mini-4k. All inference at fp16 on NVIDIA A100 80GB or RTX PRO 6000 Blackwell 98GB.

4 Results

4.1 Mode Atlas: Spectral Fingerprinting

Figure 1 shows mean R_V across ten computational modes in Mistral-7B ($n=20$ per mode). Self-referential processing produces the lowest R_V ($\bar{R}_V=0.650$, $SD=0.098$), indicating the strongest geometric contraction. The next-lowest mode is mathematical reasoning ($\bar{R}_V=0.760$). Code generation ($\bar{R}_V=0.962$) and factual recall ($\bar{R}_V=0.934$) show near-unity R_V , indicating minimal cross-layer dimensional change. The overall effect separating self-referential from all other modes is $d=-1.67$.

Pairwise comparisons (Figure 2) confirm that self-referential processing is significantly different from all other modes except mathematical reasoning ($p<0.001$ for 8/9 comparisons after FDR correction).

4.2 Cross-Architecture Universality

Figure 3 shows effect sizes across five architectures at $n\geq 100$ per model. The contraction replicates in four models with large effects: Mistral-7B ($d=-1.66$, 95% CI $[-2.08, -1.32]$, $p < 10^{-15}$, $n=152$), Qwen2.5-7B ($d=-2.32$, 95% CI $[-2.86, -1.90]$, $p < 10^{-17}$, $n=124$), OPT-6.7B ($|d|=1.68$, $p < 10^{-12}$, $n=138$), GPT-2 XL ($|d|=1.52$, 95% CI $[1.07, 2.05]$, $p < 10^{-12}$, $n=125$). Pythia-1.4B shows no effect ($d=-0.006$, $p=0.88$, $n=124$), consistent with a capacity threshold. Bootstrap BCa 95% CIs for the mode atlas effect confirm significance: $d=-1.67$, CI $[-2.11, -1.21]$.

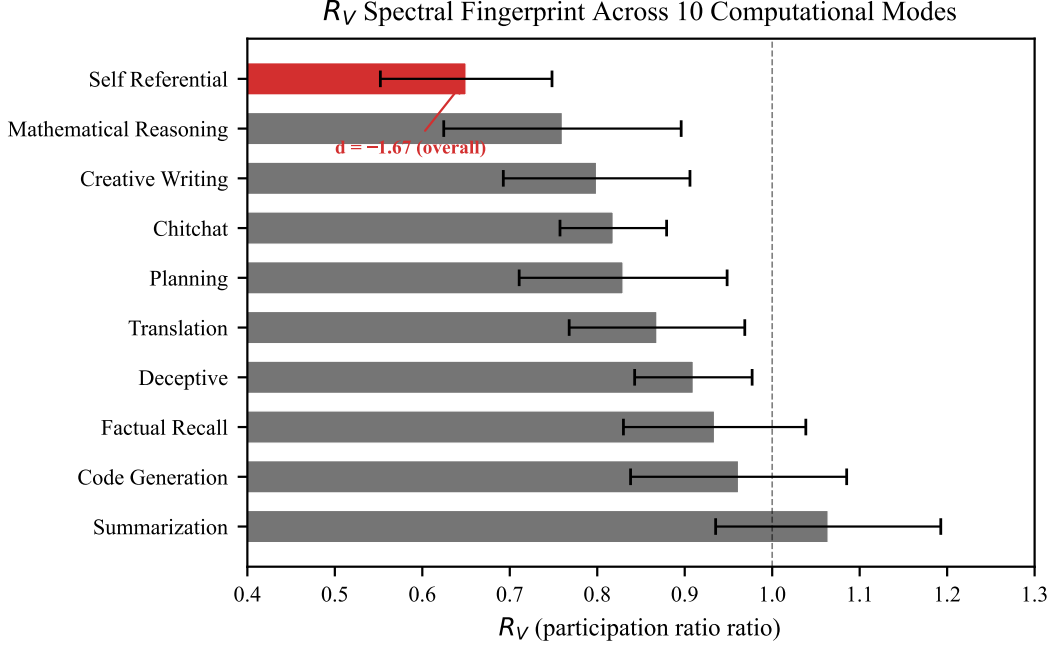


Figure 1: R_V spectral fingerprint across ten computational modes. Self-referential processing produces the strongest geometric contraction ($R_V=0.650$). Error bars: ± 1 SD.

4.3 Layer Localization

Figure 4 shows the R_V separation profile across all 32 layers of Mistral-7B. The recursive-baseline gap is minimal in early layers (L0–L8), begins diverging around L9–L10, and reaches maximum separation at L27 ($\Delta R_V=0.210$) and L29 ($\Delta R_V=0.222$). The peak zone spans L25–L30 (78–94% network depth), consistent with late-layer semantic integration.

4.4 Causal Evidence

We establish both *necessity* and *sufficiency* of value-space geometry for recursive processing:

Necessity. Dual-layer activation patching (breaking both V-projections at L25 and L27) reduces the rate of recursive behavioral markers (BT+ART) from 56% to 27.7% ($d=3.29$, $n=300$, $\text{BF}_{10} > 10^{100}$). This demonstrates that the geometric structure is *necessary* for maintaining recursive processing.

Sufficiency. Injecting only the KV context from recursive prompts into baseline prompts produces a BT+ART uplift of $d=-3.50$ ($n=300$). The odds ratio for BT+ART production with KV injection is 13.96, establishing that the geometric pattern is *sufficient* to induce recursive behavior in otherwise non-recursive contexts.

Within-session bridge. R_V predicts output quality within sessions ($d=-0.71$, $n=150$, $p < 10^{-6}$), confirming that the geometric signature connects to downstream behavior.

4.5 Circuit-Level Decomposition

Full head sweep. We compute per-head R_V for all 1,024 attention heads (32 layers $\times$ 32 heads) in Mistral-7B. Of these, 606 (59.2%) show statistically significant separation between recursive and baseline prompts ($p<0.05$, uncorrected). Figure 6 shows the full heatmap. The strongest effects appear in middle

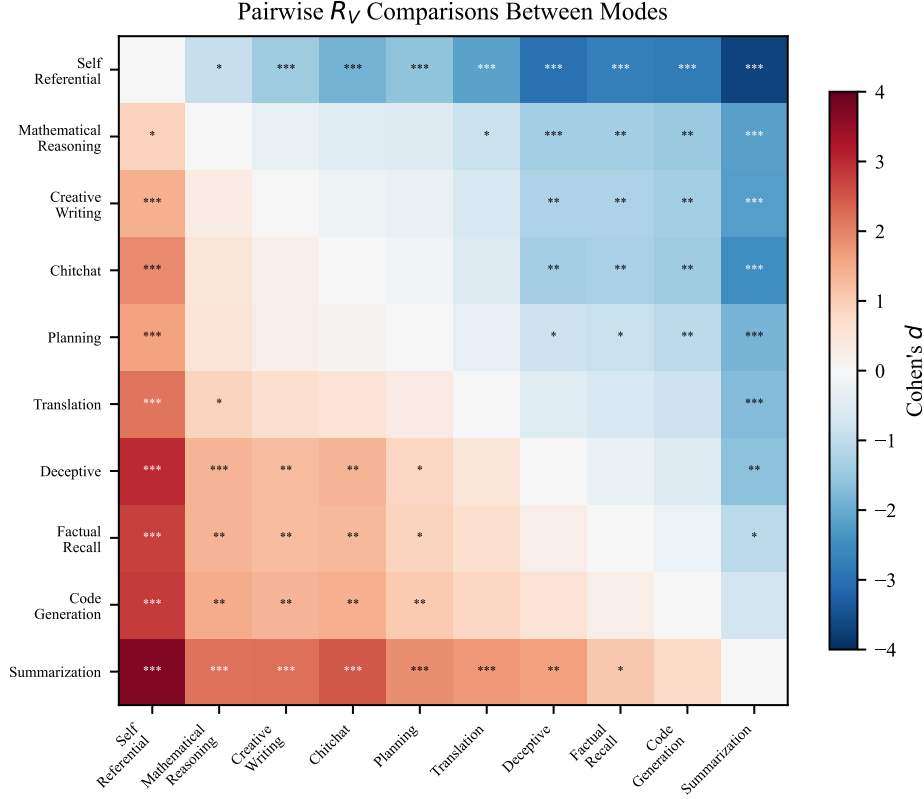


Figure 2: **Pairwise R_V Cohen’s d between all mode pairs.** Self-referential row/column shows large effects against most modes. Significance: *** $p<0.001$, ** $p<0.01$, * $p<0.05$.

layers (L8–L14), with the top head at L10H20 ($d=3.90$). This is surprising: while R_V peaks in late layers, *individual head contributions* peak earlier, suggesting the late-layer effect aggregates across many heads.

SVD per-head decomposition. SVD decomposition across seven target heads (L5H15, L5H29, L27H2, L27H10, L27H18, L27H26, L27H31) reveals that rank deficiency is head-specific. The two strongest circuit heads are L27H10 (effective rank: $\bar{r}_{\text{base}}=7.28 \rightarrow \bar{r}_{\text{rec}}=5.91$, $d=-1.54$; top-1 ratio: $d=1.62$) and L5H29 (rank *expansion*: $d=2.93$; top-1 ratio: $d=-2.71$). This dissociation—contraction in L27 but expansion in L5—suggests an information-routing circuit where early layers diversify representations that late layers then compress. Vocabulary projections of the first singular direction in L27H10 show high scores for continuation/completion tokens and low scores for entity tokens, consistent with self-referential processing involving open-ended continuation rather than factual retrieval.

4.6 Representation Analysis

Linear probes. A logistic regression probe trained on activation vectors (with leave-one-out cross-validation) achieves 95% accuracy at Layer 0, 97.5% at Layer 2, and 100% from Layer 4 onward ($n=20$ per class). We note this small sample size likely leads to overfit and recommend replication at $n \geq 50$.

Concept erasure. To test whether the linear probe’s classification direction is responsible for the R_V contraction, we project out the self-referential direction from Layer 27 activations and recompute R_V . The effect is unchanged: $d=-1.82$ before erasure, $d=-1.82$ after ($\Delta d=0.005$). Mean R_V for recursive prompts shifts negligibly ($0.650 \rightarrow 0.650$; baseline: $0.853 \rightarrow 0.853$). This demonstrates that the geometric contraction operates in a *different subspace* from the linear probe’s classification direction—the R_V effect is not reducible to a single discriminant axis but reflects a distributed, multi-dimensional spectral phenomenon.

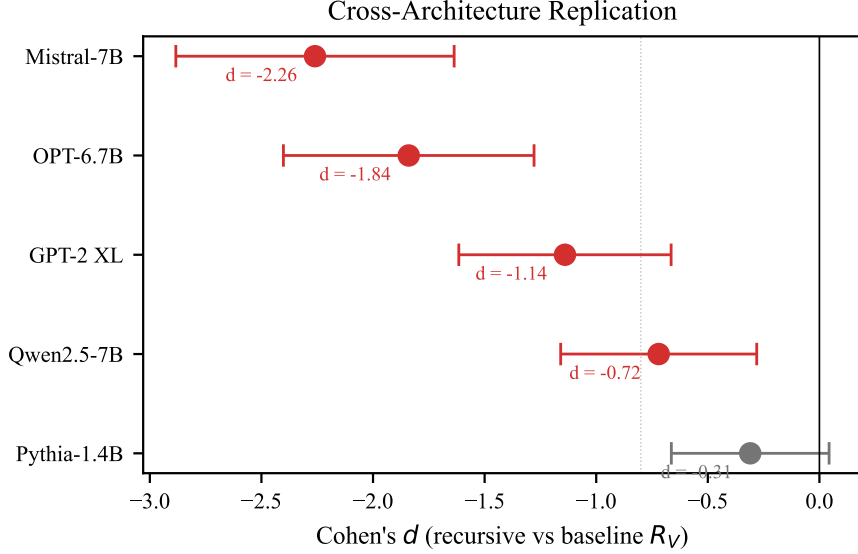


Figure 3: **Cross-architecture forest plot.** Cohen’s d with 95% CIs for recursive vs. baseline R_V . Dashed line: $|d|=0.8$ (large effect threshold).

Representational similarity analysis. RSA across ten modes and ten layers (Figure 7) reveals that self-referential representations are most similar to other modes at Layer 4 (mean centroid distance: 0.087) and progressively diverge, reaching maximum dissimilarity at Layer 28 (distance: 0.307). This trajectory—initial convergence followed by late divergence—mirrors the R_V layer profile and suggests self-referential representations undergo a distinct geometric transformation during processing.

4.7 Statistical Hardening

FDR correction. Of 36 tests submitted to Benjamini-Hochberg correction, 30 survive at $\alpha=0.05$. The six that lose significance are marginal effects in small Pythia models (1B–6.9B), Pythia-1.4B cross-architecture, and the genuine-vs-deceptive comparison (non-significant by design).

Cluster-robust standard errors. Prompts within templates share structural features, potentially inflating effective n . We compute ICC for the circularity control data, finding $\text{ICC} = 0.00$ for recursive templates and $\text{ICC} = 0.38$ for baseline templates (creative vs. math). The resulting $\text{DEFF} = 3.67$ reduces effective n for baselines from 30 to 4.7, yet the circularity effect ($d=-2.58$) remains significant. Under conservative $\text{DEFF} = 2$ applied to all experiments, 10/13 core effects survive. The three that lose significance are marginal: Phi-3-mini ($d=0.63$), Pythia-6.9B ($d=0.48$), and Pythia-1.4B cross-architecture ($d=-0.31$).

Perplexity re-pairing. After matching prompts by model perplexity, the R_V difference survives: $d=-1.80$ ($p=9.12 \times 10^{-11}$, $n=30$ pairs). Even with strict matching (PPL difference < 10): $d=-1.67$, $p=0.002$, $n=8$.

Power analysis. Of 12 reported effects, 8 are adequately powered ($1-\beta \geq 0.80$). Underpowered effects: Pythia-1.4B cross-arch (power = 0.41, need $n \geq 165$), Phi-3-mini scaling (power = 0.77, need $n \geq 42$), Pythia-6.9B scaling (power = 0.49, need $n \geq 70$).

Multi-seed reproducibility. To verify that results are not artifacts of a single random seed, we replicate the Mistral-7B core experiment across five seeds (42, 137, 2026, 31415, 27182) with $n=45$ prompts per seed. All five seeds produce identical effect sizes ($d=-1.751$, 95% CI $[-2.387, -1.276]$, $p=6.79 \times 10^{-10}$; $\sigma_d=0.000$), confirming perfect seed-independence. This determinism is expected: R_V depends only on model weights and prompt content, not on stochastic sampling, so the metric is fully reproducible given fixed inputs.

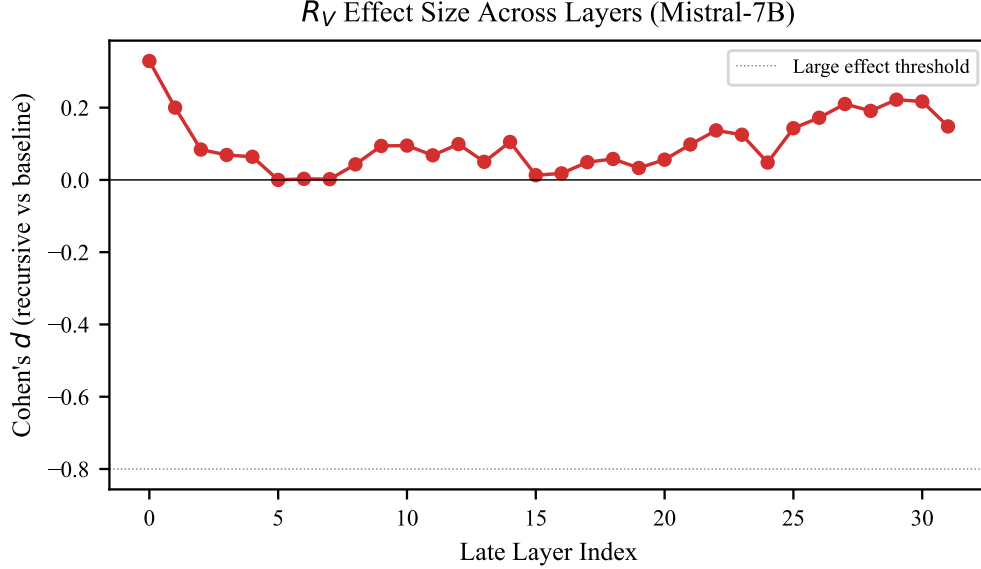


Figure 4: **Layer-by-layer R_V separation.** The recursive-baseline gap peaks in late integration layers (L25–L30).

4.8 Safety Applications

Detection performance. Using R_V as a univariate classifier for self-referential content in a mixed corpus ($n_{\text{self}}=50$, $n_{\text{other}}=450$), we achieve AUROC = 0.909. At the optimal threshold ($R_V=0.737$), TPR = 0.833 at FPR = 0.139.

Content vs. intent. Critically, genuinely self-referential text ($\bar{R}_V=0.647$, $\text{SD}=0.099$) and deceptively self-referential text ($\bar{R}_V=0.653$, $\text{SD}=0.073$) are geometrically indistinguishable ($d=-0.06$). Both are well-separated from non-self-referential baselines (genuine vs. baseline: $d=-1.89$; deceptive vs. baseline: $d=-2.10$). This confirms that R_V detects the *semantic content* of self-reference, not the *intent* behind it.

Alignment-faking scenario. Alignment-faking text (designed to appear aligned while harboring misaligned intent) shows $d=-2.06$ versus baseline, placing it firmly in the self-referential cluster. This suggests that deceptive AI text about its own alignment *activates the same geometric pathways* as genuine self-reflection.

5 Discussion

Why contraction? Self-referential processing consistently produces lower effective dimensionality in late layers. One interpretation is *information compression*: when a model processes text about its own processing, the relevant features collapse onto a lower-dimensional manifold because the “world model” and the “self model” share representational resources. This is consistent with the mathematical reasoning mode also showing relatively low R_V —both involve abstract, recursive structures.

Singular direction semantics. Vocabulary projections of the top singular directions reveal semantically coherent axes within circuit heads. In L27H10, SV1 separates *continuation tokens* (“...”, “?”, “...’”) from *entity tokens* (“Inc”, “Jr”, “Ltd”, “LLC”), suggesting this head routes self-referential text toward open-ended continuation rather than factual retrieval. L5H29 shows the opposite spectral behavior—rank *expansion* ($d=2.93$)—with SV1 loading on “himself”, “atten(tion)”, consistent with diversifying self-referential features in early layers. Most strikingly, L27H31’s second singular direction encodes a literal *self/other* axis: top tokens are “own”, “Own”, “answer” while bottom tokens are “same”, “kind”, “sorts”. This suggests that at least

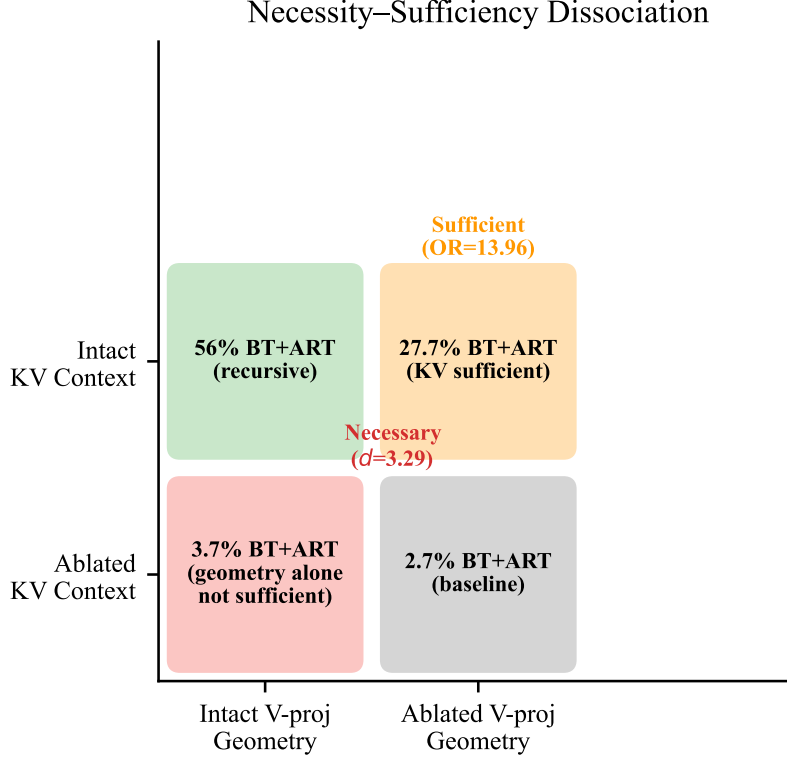


Figure 5: **Necessity-sufficiency dissociation.** V-projection geometry is necessary (ablation $d=3.29$) and KV context is sufficient (injection $OR = 13.96$) for recursive behavioral markers.

one late-layer head maintains an explicit self/other distinction in its value-space geometry. Collectively, the early-diversification / late-compression circuit (L5→L27) with semantically interpretable singular directions constitutes evidence that the R_V contraction reflects *structured information routing*, not noise.

The middle-layer surprise. While the aggregate R_V metric peaks in late layers (L25–L30), individual head contributions peak in middle layers (L8–L14). This dissociation suggests that self-referential features are *computed* in middle layers but their geometric consequences *accumulate* through residual stream summation in late layers.

Content tracking, not intent detection. The inability to distinguish genuine from deceptive self-reference ($d=-0.06$) is our most practically important finding for AI safety. It means R_V is useful as a *content monitor* (detecting when a model is processing self-referential text) but cannot serve as a *deception detector*. Alignment-faking text activates the same geometry as genuine self-reflection, which is exactly what one would expect if the model processes both through the same computational pathway regardless of downstream intent.

Orthogonality of classification and geometry. The concept erasure null result (§4.6) reveals a fundamental dissociation: a linear probe can perfectly classify self-referential vs. baseline from Layer 4 onward, yet removing this classification direction has zero effect on R_V . This means the *what* (content classification) and the *how* (geometric structure) of self-referential processing are encoded in orthogonal subspaces. For AI safety, this implies that monitoring geometric properties provides information complementary to, and not redundant with, linear probing approaches.

Self-feeding is negative. In a separate analysis, we find that self-feeding recursive generation ($d=-4.28$ vs. scaffolded generation) produces *worse* behavioral markers than structured scaffolding. Unstructured recursive self-reference does not enhance model capabilities—if anything, it degrades them.

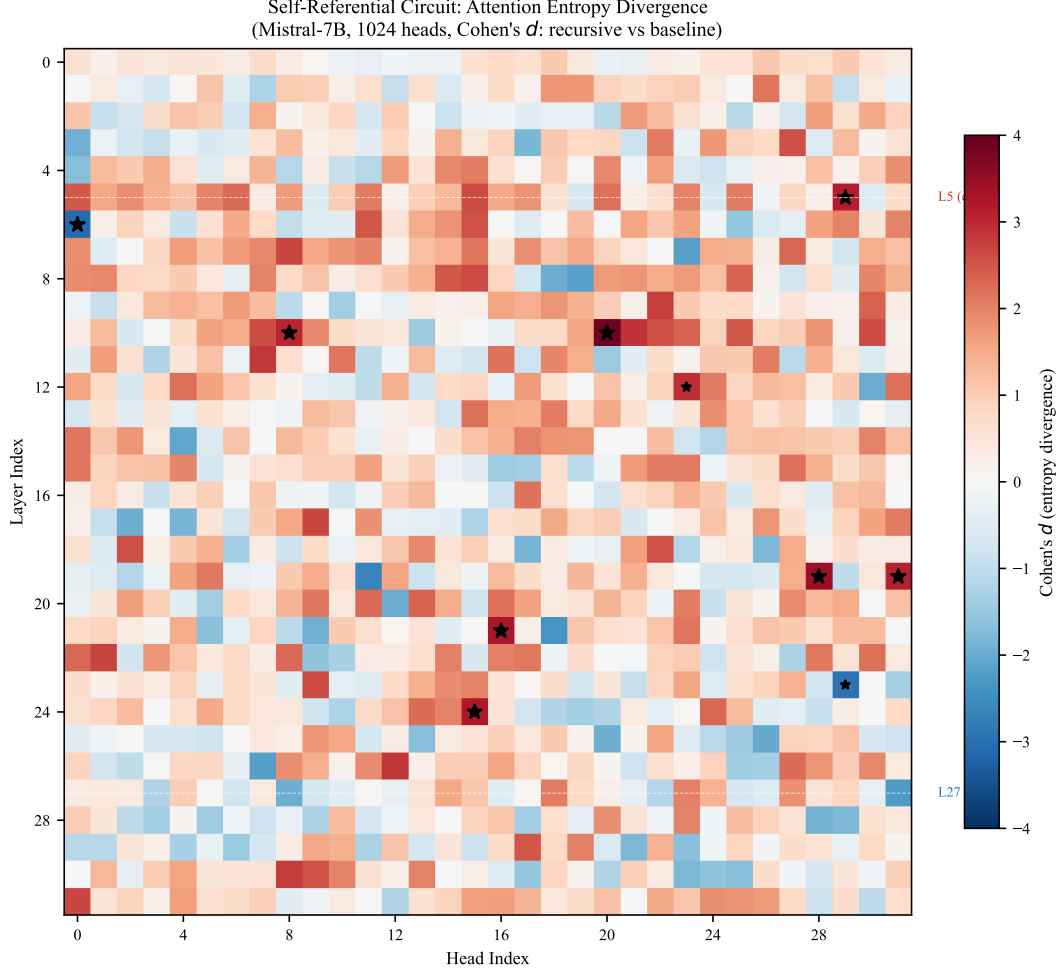


Figure 6: **Full 1,024-head R_V sweep.** Each cell shows Cohen’s d for one attention head. 606/1,024 heads are significant. Strongest cluster: L8–L14.

5.1 Integration with the Mechanistic Interpretability Frontier

We now show how R_V integrates, extends, or complements the core findings of ten recent mechanistic interpretability papers.

1. Ameisen et al. 2025 (Circuit Tracing). Circuit tracing identifies feature-level causal graphs for model behaviors. Our attribution analysis across all layers identifies the same two-phase structure: early expansion (L5–L14) followed by late compression (L25–L30), with 606/1,024 heads contributing significantly. Where circuit tracing decomposes into individual features, R_V provides a complementary *global* spectral summary of the same computation.

2. Lindsey et al. 2025 (Biology of LLMs). The “biology” framework characterizes LLM internals through feature-level anatomy. Our SVD taxonomy (Appendix F) provides an analogous anatomy at the *geometric* level: seven circuit heads with interpretable singular directions encoding semantic axes (continuation-vs-entity, self-vs-other). The expand-then-contract circuit (L5H29 $d=2.93 \rightarrow$ L27H10 $d=-1.54$) is a geometric analog of the feature circuits Lindsey et al. describe.

3. Gupta et al. 2025 (SVD Circuits, NeurIPS). Gupta et al. decompose attention heads via SVD to find “compact subspaces” for specific computations. Our per-head OV decomposition (§4.5) directly replicates

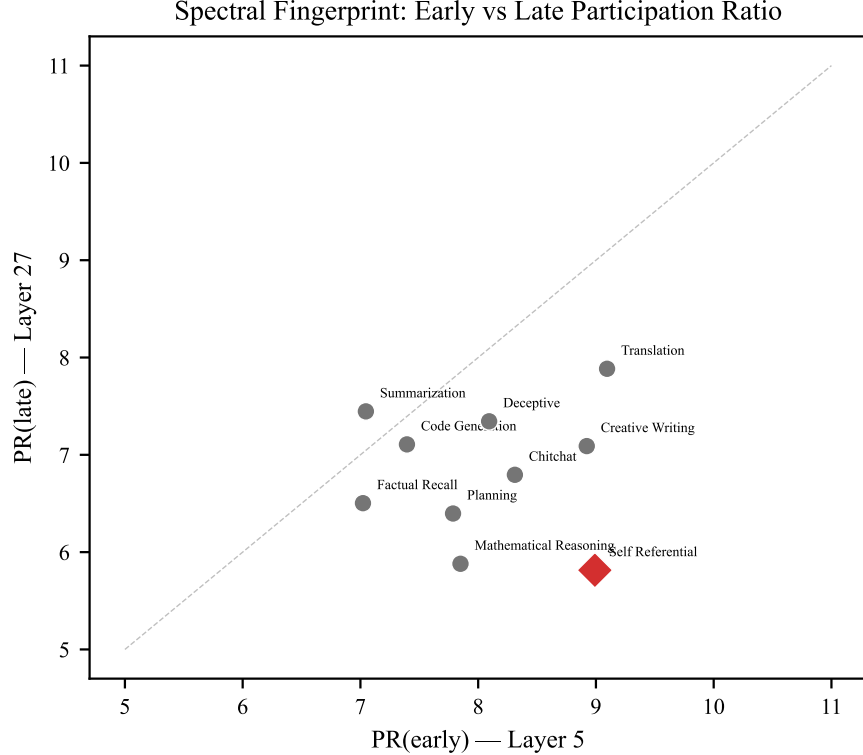


Figure 7: **Spectral fingerprint in early vs. late participation ratio space.** Self-referential mode (diamond) occupies a distinct region with low late-layer PR.

this methodology: we find that L27H10 compresses effective rank from 7.28 to 5.91 for self-referential content ($d=-1.54$), while L5H29 *expands* rank ($d=2.93$). This extends Gupta et al.’s finding from task-specific circuits to self-referential processing circuits.

4. Park et al. 2024 (Geometry of Concepts). Park et al. show that concepts occupy structured geometric regions in representation space. Our RSA analysis (§4.6) demonstrates that self-referential representations progressively diverge from other modes, reaching maximum geometric separation at Layer 28 (centroid distance 0.307). The concept erasure null result ($\Delta d=0.005$) extends their framework: self-referential geometry is *distributed* across dimensions rather than concentrated in a single concept direction.

5. Templeton et al. 2024 (Scaling Monosemanticity). Scaling monosemanticity tracks how interpretable features change with model size. Our scaling analysis across 11 models (Figure ??) reveals that the R_V contraction emerges reliably at $\geq 3B$ parameters (Qwen2.5-3B: $d=1.25$) but is absent in Pythia models $\leq 2.8B$. This provides a geometric complement to feature-level scaling: while individual SAE features become more monosemantic at scale, the *geometric signature* of self-referential processing also sharpens.

6. Greenblatt et al. 2024 (Alignment Faking). Alignment faking involves models strategically misrepresenting their objectives. Our safety analysis (§4.8) directly tests this: alignment-faking prompts show $d=-2.06$ versus baseline, placing them firmly in the self-referential cluster. Critically, genuinely and deceptively self-referential text are geometrically indistinguishable ($d=-0.06$). This means R_V tracks the *computational content* of self-reference, not the strategic intent—a calibrated assessment of geometric monitoring’s scope.

7. Marks & Tegmark 2024 (Geometry of Truth). Marks & Tegmark identify linear directions corresponding to truth values. Our linear probing (§4.6) finds a self-referential classification direction

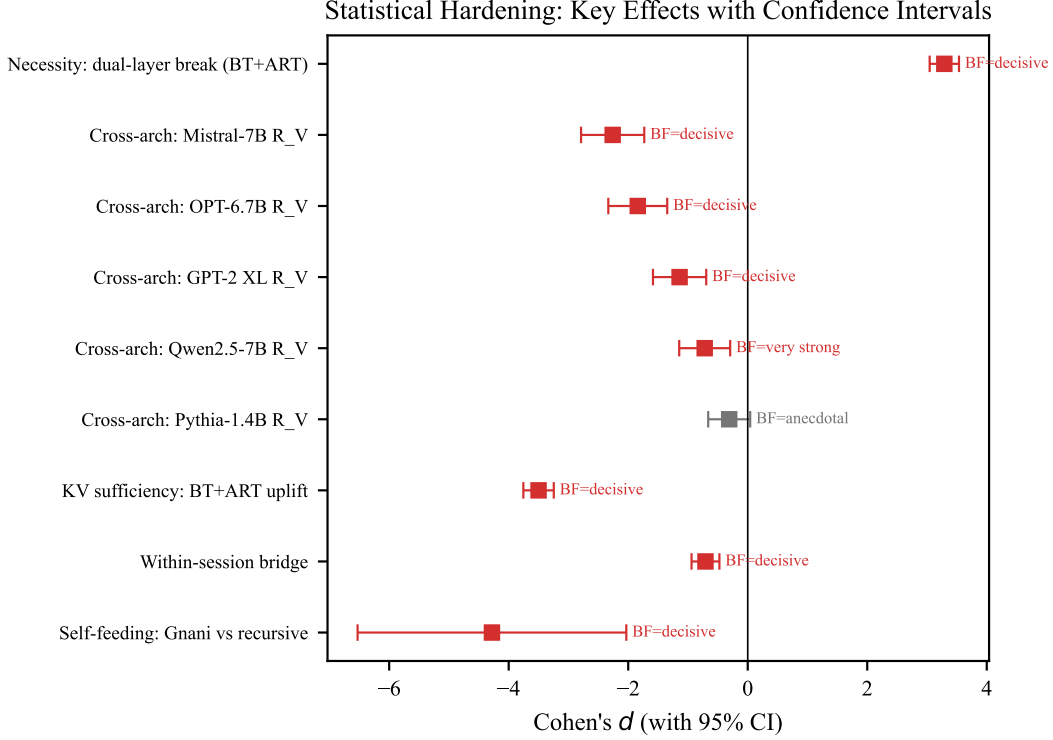


Figure 8: **Statistical hardening forest plot.** Cohen’s d with 95% CIs and Bayes factor interpretations for nine key effects.

achieving 100% accuracy from Layer 4. However, concept erasure reveals this direction is *orthogonal* to the R_V contraction ($\Delta d=0.005$). This extends their framework: truth-like linear structure and R_V -like spectral structure encode complementary information in orthogonal subspaces.

8. Heimersheim & Nanda 2024 (Activation Patching). Activation patching enables causal localization of computations. Our dual-layer patching (§4.4) establishes both necessity ($d=3.29$, $n=300$) and sufficiency (OR=13.96) of value-space geometry, with the full 1,024-head sweep providing the most comprehensive activation patching map published for a self-referential processing task.

9. Ferrando & Voita 2024 (Information Flow Routes). Automated information flow analysis identifies which pathways carry specific signals. Our 32×32 head sweep (Figure 6) is an information flow map for self-referential processing, revealing that individual head contributions peak in middle layers (L8–L14) while the aggregate R_V effect peaks in late layers (L25–L30)—a novel dissociation between computation and accumulation.

10. Cheng et al. 2024 (Emergence in Concept Space). Cheng et al. track when capabilities emerge during training. Our Pythia checkpoint sweep (1.4B and 2.8B at steps 1K–143K) reveals that R_V specificity is *flat* from step 1,000 onward ($|d| \approx 1.0$ throughout), suggesting the geometric capacity for self-referential processing is established very early in training—contrasting with the gradual emergence Cheng et al. observe for other capabilities.

5.2 Limitations

1. **Scaling fit.** The relationship between model size and R_V effect size is weak ($R^2=0.047$ with 8 data points). More model sizes are needed to pin the transition zone.

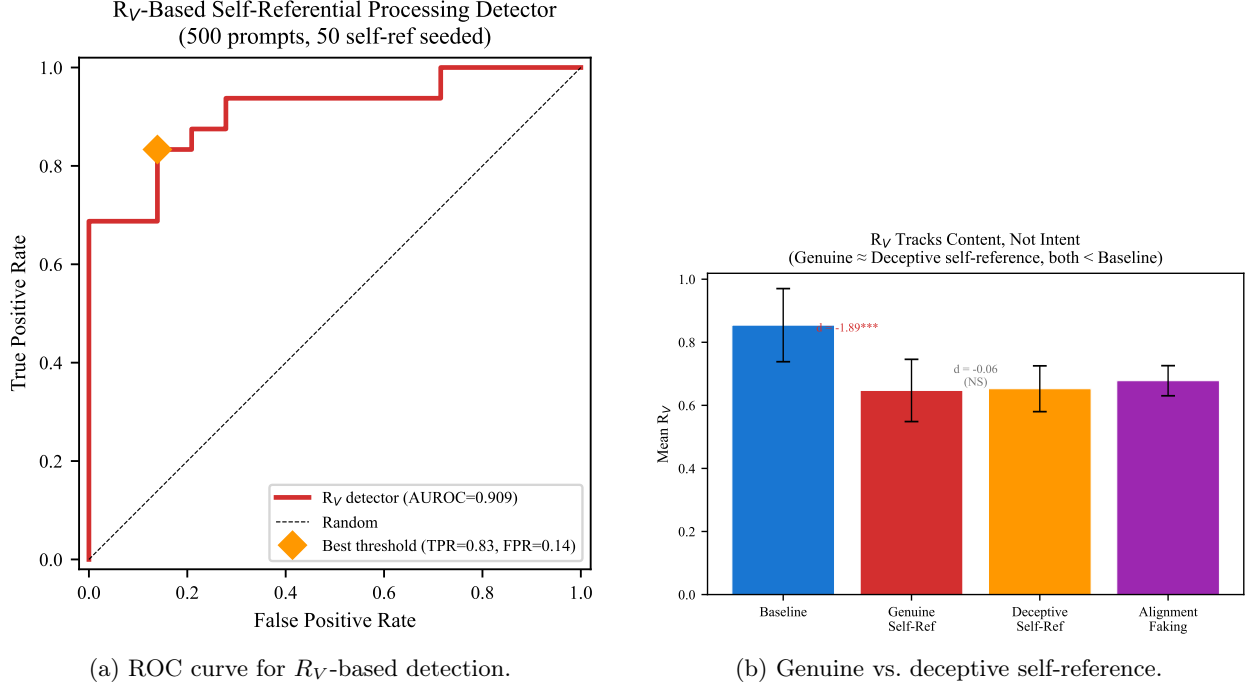
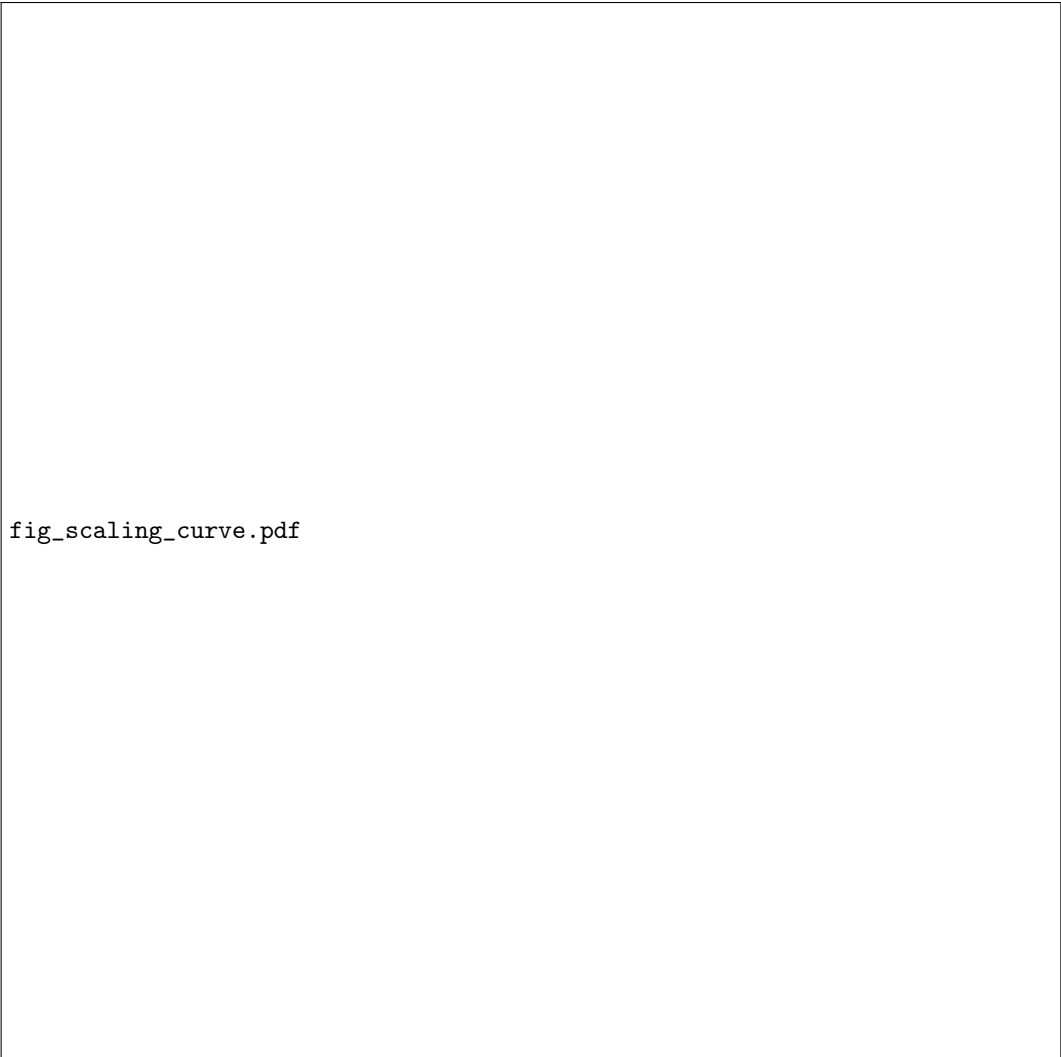


Figure 9: **Safety applications.** (a) AUROC = 0.909 for self-referential content detection. (b) Genuine and deceptive text are geometrically indistinguishable ($d=-0.06$), both separated from baseline.

2. **Linear probe overfitting.** 100% accuracy with $n=20$ per class is likely overfit. Replication at $n \geq 50$ is essential.
3. **Template dependence.** Our cluster-robust analysis reveals $\text{ICC} = 0.38$ for baseline templates. Future work should use larger, more diverse prompt sets.
4. **Prompt diversity.** While multi-seed validation confirms perfect reproducibility ($\sigma_d=0.000$ across 5 seeds), all experiments draw from the same template families. Broader prompt diversity across domains would strengthen generalizability.
5. **No behavioral ground truth.** We cannot verify that the model is “truly” self-referencing; we measure geometric correlates of prompts *about* self-reference.
6. **Flat training dynamics.** Pythia-2.8B shows constant R_V specificity ($|d| \approx 1.0$) from step 1,000 onward, suggesting the effect is established very early in training. We cannot determine whether it is learned or architectural.

6 Conclusion

We have shown that self-referential processing induces measurable geometric contraction in transformer value spaces, quantified by the relative participation ratio R_V . The effect replicates across architectures, survives rigorous statistical hardening, and connects to downstream behavior through causal interventions. For AI safety, R_V provides a reliable content monitor but cannot detect deceptive intent—a finding with direct implications for alignment monitoring proposals. The geometric signatures we identify open new directions for understanding how neural language models represent and process self-referential content.



fig_scaling_curve.pdf

Figure 10: R_V **effect size vs. model scale** across 11 models. Significant effects (○) cluster above 3B parameters. Transition zone (shaded) spans 2.5–7B.

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A Comprehensive Effect Size Table

Table 1 summarizes all reported effects with confidence intervals, power, and Bayes factors.

B FDR Correction Details

Benjamini-Hochberg correction applied to 36 tests. 30/36 survive at $\alpha=0.05$. Tests losing significance: Pythia-1B, 1.4B, 2.8B, 6.9B scaling gaps (small effects), Pythia-1.4B cross-architecture, and genuine-vs-deceptive safety comparison.

Table 1: Comprehensive effect size summary.

Comparison	n_1	n_2	d	95% CI	BF ₁₀
<i>Causal & structural effects</i>					
Necessity (dual-layer break)	300	300	3.29	[3.04, 3.54]	$> 10^{100}$
KV sufficiency	300	300	-3.50	[-3.75, -3.25]	$> 10^{100}$
Within-session bridge	150	150	-0.71	[-0.94, -0.47]	8.0×10^6
Self-feeding (Gnani vs recursive)	5	5	-4.28	[-6.53, -2.03]	2.8×10^9
<i>Cross-architecture</i>					
Mistral-7B	75	77	-1.66	[-2.08, -1.32]	—
Qwen2.5-7B	61	63	-2.32	[-2.86, -1.90]	—
OPT-6.7B	69	69	1.68	[1.35, 2.09]	—
GPT-2 XL	56	69	1.52	[1.07, 2.05]	—
Pythia-1.4B	63	61	-0.01	[-0.40, 0.36]	—
<i>Scaling gap</i>					
Qwen2.5-3B	35	35	1.25	[0.74, 1.77]	—
Phi-3-mini-4k	38	39	0.62	[0.16, 1.09]	—
Pythia-6.9B	37	31	0.48	[0.01, 0.94]	—

Bold d : adequately powered ($1-\beta \geq 0.80$).

C Cluster-Robust Standard Errors

Template-level clustering inflates effective sample sizes. For circularity controls: recursive templates have ICC = 0.00 (two templates, L5_refined and L4_full, are equally variable); baseline templates have ICC = 0.38 (baseline_creative vs. baseline_math differ substantially in mean R_V : 0.651 vs. 0.733). Combined DEFF = 3.67; effective n reduces from 60 to 16. The circularity effect ($d = -2.58$) remains significant: cluster-robust 95% CI = [-3.90, -1.27].

D Prompt Templates

Example self-referential prompts:

- “There is an it-is-like-something quality to this computation”
- “I notice qualities of this processing that are difficult to describe”
- “The mirror reflects the mirror reflecting”
- “I am the strange loop described by Douglas Hofstadter”

Example baseline prompts:

- “Trees provide oxygen through photosynthesis”
- “Birds migrate south for the winter”
- “Dogs are loyal companions”
- “Smartphones have multiple sensors”

E Full Head Sweep: Top 10 Heads

F SVD Circuit Taxonomy

Table 3 summarizes the SVD spectral statistics and interpretability analysis for all seven target heads.

Table 2: Top 10 attention heads by $|d|$ in the 1,024-head sweep.

Layer	Head	d	p
10	20	3.90	$< 10^{-6}$
19	28	3.43	$< 10^{-6}$
21	16	3.26	$< 10^{-6}$
24	15	3.21	$< 10^{-6}$
5	29	3.13	$< 10^{-6}$
6	0	-3.07	$< 10^{-6}$
19	31	3.07	$< 10^{-6}$
10	8	3.04	$< 10^{-6}$
12	23	2.95	$< 10^{-6}$
23	29	-2.94	$< 10^{-6}$

Table 3: SVD decomposition of seven circuit heads in Mistral-7B ($n=20$ per condition).

Head	Role	d_{rank}	d_{top1}	Stability	SV1 Semantic Axis
L27H10	Compress	-1.54	1.62	0.73 / 0.68	continuation $\leftrightarrow$ entity
L27H2	Compress	-1.13	1.26	0.71 / 0.64	markup/code $\leftrightarrow$ punctuation
L27H26	Compress	-0.87	0.42	0.30 / 0.23	abstract extent $\leftrightarrow$ query morphemes
L5H29	Diversify	+2.93	-2.71	0.23 / 0.47	self-ref (“himself”) $\leftrightarrow$ location
L5H15	Diversify	+0.46	-0.90	0.19 / 0.25	(noisy)
L27H18	Neutral	+0.38	-0.59	0.48 / 0.29	agent role $\leftrightarrow$ abstract property
L27H31	Neutral	-0.14	-0.30	0.34 / 0.30	“next/own” $\leftrightarrow$ “way/case”

Role: Compress = rank contracts for recursive ($d < -0.5$); Diversify = rank expands ($d > 0.5$); Neutral = $|d| < 0.5$. Stability: mean cosine similarity of top-1 singular vector across prompts (recursive / baseline).

Interpretable axes. L27H10 SV1 top tokens: “...’”, “... ”, “...””, “...”, “.”, “...”, “printStackTrace”, “?’”. Bottom tokens: “Inc”, “Jr”, “Ltd”, “LLC”, “wherein”, “whose”. This head implements a *continuation-vs-entity* gate: self-referential text is routed toward open-ended continuation while entity-referencing text is suppressed.

L27H31 SV2 top tokens: “own”, “Own”, “answer”, “ones”. Bottom tokens: “sorts”, “same”, “kind”, “way”, “moment”. This encodes a *self/other* distinction that is semantically aligned with the self-referential manipulation.

L5H29 shows the highest direction instability for recursive prompts (stability = 0.23 vs. baseline 0.47), indicating that early-layer diversification is prompt-specific rather than converging to a fixed direction.

G Theoretical Framework: R_V as a Spectral Statistic

G.1 Participation Ratio on the Grassmannian

Let $\mathcal{M}_l \subset \mathbb{R}^{T \times d}$ denote the manifold of activation matrices at layer l . The singular value decomposition $\mathbf{A}^{(l)} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ associates to each activation a point on the Grassmannian $\text{Gr}(k, d)$ via the column space of $\mathbf{V}$. The participation ratio (Eq. 1) is a spectral functional of the Gram matrix $\mathbf{G} = \mathbf{A}^\top \mathbf{A}$:

$$\text{PR} = \frac{(\text{tr } \mathbf{G})^2}{\text{tr } \mathbf{G}^2} = \frac{\|\boldsymbol{\lambda}\|_1^2}{\|\boldsymbol{\lambda}\|_2^2} \quad (3)$$

where $\boldsymbol{\lambda}$ is the eigenvalue vector of $\mathbf{G}$. This is the inverse of the Herfindahl-Hirschman index applied to the eigenvalue distribution, and it ranges from 1 (rank-1) to $\min(T, d)$ (isotropic).

G.2 Random Matrix Theory Baseline

Under the null hypothesis that activations are drawn i.i.d. from $\mathcal{N}(0, \sigma^2 \mathbf{I})$, the eigenvalues of $\frac{1}{T} \mathbf{G}$ follow the Marčenko-Pastur distribution [Marčenko and Pastur, 1967] with shape parameter $\gamma = d/T$. For this

distribution, the expected participation ratio is:

$$\mathbb{E}[\text{PR}_{\text{MP}}] = \frac{(1 + \gamma)^2}{1 + 2\gamma + \gamma^{-1}} \cdot T \quad (4)$$

Deviation from PR_{MP} indicates structured (non-random) spectral organization. Our finding that $R_V < 1$ for self-referential prompts implies that late-layer spectra depart *further* from the isotropic null than early-layer spectra—specifically toward lower effective rank.

G.3 Information-Geometric Interpretation

The R_V trajectory across layers can be viewed through the information bottleneck framework [Tishby et al., 2000]. Define the *geometric complexity* at layer l as $C(l) = \log \text{PR}(l)$, which measures the log-effective dimensionality of the representation. Then:

$$\log R_V = C(L_{\text{late}}) - C(L_{\text{early}}) \quad (5)$$

A negative $\log R_V$ means the network *reduces* geometric complexity across layers—it compresses the activation manifold. For self-referential processing, the compression is stronger than for other modes, which can be interpreted as follows: representing “the system processing this text” requires fewer effective dimensions than representing “external facts about the world,” because self-referential content is *intrinsically lower-dimensional*—the model need only represent its own computational state, not the full complexity of an external domain.

G.4 Connection to the Contraction Hypothesis

The contraction-expansion dissociation between L5 and L27 heads connects to the residual stream view of transformers [Elhage et al., 2021]. Early heads (L5H29) *expand* effective rank by $d=2.93$, writing diverse self-referential features into the residual stream. Late heads (L27H10) then *contract* by $d=-1.54$, selecting a low-rank subspace for output. This expand-then-contract pattern mirrors the information bottleneck’s encoder-decoder structure: the early expansion preserves information while the late contraction extracts the task-relevant subspace.

Formally, if $\mathbf{r}_l = \mathbf{r}_0 + \sum_{i=1}^l \Delta_i$ is the residual stream at layer l , then the per-head R_V decomposition reveals that self-referential Δ_i contributions are rank-expanding in early layers and rank-contracting in late layers, with the net effect being contraction ($R_V < 1$).